

Configuring a multi-area OSPF network

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Purpose

This Lab acts as a refresher of memory for a widely used routing protocol, OSPF, including its commands and concepts for ipv4 and ipv6, as well as introducing the students what kind of labs they would be performing in the near future.

Background Information/Lab Concepts

OSPF:

OSPF, or Open shortest path first is a routing protocol used by many. It's an Interior Gateway Protocol that's commonly used in large enterprise networks. OSPF is based on the Shortest Path First algorithm, which uses graph theory to find the shortest path between nodes in a network.

Layer 3 Switch:

A Layer 3 switch is different from a layer 2 switch, as it combines the functions of a switch and a router. Layer 3 switches are neat in that they are much more compact, as it combines both the router and a switch into one instead of having a router and a switch in place. It is also easy to configure each port for different uses based off of needs.

Lab Summary

First, students had to create a diagram/guide of the lab as a simulation in packet tracer, then create an address table for all of the equipment that will be used within the lab and assign each piece of equipment to a unique address (ipv4 and ipv6). Afterwards, they would replicate the simulated scenario with physical routers and switch, connecting each device with ethernet cords, and consoling into them, configuring their ip address and setting up OSPF routing protocol between routers and switches. When they are finished, they would test the connectivity between devices and if any error occurs, troubleshoot those errors. All traffic between routers and switches are meant to be routed through OSPF.

Lab Commands

(Unnecessary or extra lines, such as “!” are removed to save space)

R1:

Building configuration...

Current configuration : 4043 bytes

! Last configuration change at 23:52:53 UTC Tue Sep 17 2024

version 16.9

service timestamps debug datetime msec

service timestamps log datetime msec

platform qfp utilization monitor load 80

platform punt-keepalive disable-kernel-core

hostname R1

boot-start-marker

boot-end-marker

vrf definition Mgmt-intf

address-family ipv4

exit-address-family

address-family ipv6

exit-address-family

no aaa new-model

login on-success log

subscriber templating

vtp domain cisco

vtp mode transparent

ipv6 unicast-routing

multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-2189345785
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-2189345785
revocation-check none
rsakeypair TP-self-signed-2189345785

!

crypto pki certificate chain TP-self-signed-2189345785
certificate self-signed 01

!

license udi pid ISR4321/K9 sn FDO21482DXE
license boot level appxk9
no license smart enable
diagnostic bootup level minimal

!

spanning-tree extend system-id

!

redundancy
mode none

!

interface GigabitEthernet0/0/0
ip address 192.168.0.3 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:1::3/64
ipv6 ospf 1 area 1

!

interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0

negotiation auto

ipv6 address 2001:DB8:ACAD:3::1/64

ipv6 ospf 1 area 1

!

interface Serial0/1/0

!

interface Serial0/1/1

!

interface GigabitEthernet0/2/0

no ip address

shutdown

negotiation auto

!

interface GigabitEthernet0/2/1

no ip address

shutdown

negotiation auto

!

interface GigabitEthernet0

vrf forwarding Mgmt-intf

no ip address

shutdown

negotiation auto

!

router ospf 1

router-id 1.1.1.1

network 192.168.0.0 0.0.0.255 area 1

network 192.168.2.0 0.0.0.255 area 1

!

ip forward-protocol nd

ip http server

ip http authentication local

ip http secure-server

ip tftp source-interface GigabitEthernet0

!

ipv6 router ospf 1

router-id 1.1.1.1

!

control-plane

!

line con 0

transport input none

stopbits 1

line aux 0

stopbits 1

line vty 0 4

login

!

End

R2:

Building configuration...

Current configuration : 3993 bytes

!

! Last configuration change at 22:49:37 UTC Tue Sep 17 2024

!
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core

!
hostname R2

!
boot-start-marker
boot-end-marker
!
vrf definition Mgmt-intf

!
address-family ipv4
exit-address-family

!
address-family ipv6
exit-address-family

!
no aaa new-model

!
login on-success log

!
subscriber templating

!
vtp domain cisco

vtp mode transparent

ipv6 unicast-routing

multilink bundle-name authenticated

!

crypto pki trustpoint TP-self-signed-2557841031

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2557841031

revocation-check none

rsakeypair TP-self-signed-2557841031

!

crypto pki certificate chain TP-self-signed-2557841031

certificate self-signed 01

!

license udi pid ISR4321/K9 sn FDO21500G1N

license boot level appxk9

no license smart enable

diagnostic bootup level minimal

!

spanning-tree extend system-id

!

redundancy

mode none

!

interface GigabitEthernet0/0/0

ip address 192.168.3.1 255.255.255.0

negotiation auto

ipv6 address 2001:DB8:ACAD:4::1/64

ipv6 ospf 1 area 0

!

interface GigabitEthernet0/0/1

ip address 192.168.2.2 255.255.255.0

negotiation auto

ipv6 address 2001:DB8:ACAD:3::2/64

ipv6 ospf 1 area 1

!

interface Serial0/1/0

!

interface Serial0/1/1

!

interface GigabitEthernet0/2/0

negotiation auto

!

interface GigabitEthernet0/2/1

negotiation auto

!

interface GigabitEthernet0

vrf forwarding Mgmt-intf

no ip address

shutdown

negotiation auto

!

router ospf 1

router-id 2.2.2.2

network 192.168.2.0 0.0.0.255 area 1

network 192.168.3.0 0.0.0.255 area 0

!

ip forward-protocol nd

ip http server

ip http authentication local

ip http secure-server

ip tftp source-interface GigabitEthernet0

!

ipv6 router ospf 1

router-id 2.2.2.2

!

control-plane

!

line con 0

transport input none

stopbits 1

line aux 0

stopbits 1

line vty 0 4

login

!

End

R3:

Building configuration...

Current configuration : 3818 bytes

!

! Last configuration change at 23:10:27 UTC Tue Sep 17 2024

!

version 16.9

service timestamps debug datetime msec

service timestamps log datetime msec

platform qfp utilization monitor load 80

platform punt-keepalive disable-kernel-core

!

hostname R3

!

boot-start-marker

boot-end-marker

!

vrf definition Mgmt-intf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

no aaa new-model

!

login on-success log

!

subscriber templating

vtp domain cisco

vtp mode transparent

ipv6 unicast-routing

multilink bundle-name authenticated

!

crypto pki trustpoint TP-self-signed-318861592

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-318861592

revocation-check none

rsa-keypair TP-self-signed-318861592

!

crypto pki certificate chain TP-self-signed-318861592

certificate self-signed 01

!

license udi pid ISR4321/K9 sn FDO214420HM

no license smart enable

diagnostic bootup level minimal

!

spanning-tree extend system-id

!

redundancy

mode none

!

interface GigabitEthernet0/0/0

ip address 192.168.3.2 255.255.255.0

negotiation auto

ipv6 address 2001:DB8:ACAD:4::2/64

ipv6 ospf 1 area 0

```
!  
interface GigabitEthernet0/0/1  
  ip address 192.168.4.1 255.255.255.0  
  negotiation auto  
  ipv6 address 2001:DB8:5::1/64  
  ipv6 ospf 1 area 2  
!  
interface Serial0/1/0  
!  
interface Serial0/1/1  
!  
interface GigabitEthernet0  
  vrf forwarding Mgmt-intf  
  no ip address  
  shutdown  
  negotiation auto  
!  
router ospf 1  
  router-id 3.3.3.3  
  network 192.168.3.0 0.0.0.255 area 0  
  network 192.168.4.0 0.0.0.255 area 2  
!  
ip forward-protocol nd  
ip http server  
ip http authentication local  
ip http secure-server  
ip tftp source-interface GigabitEthernet0
```

```
!  
ipv6 router ospf 1  
router-id 3.3.3.3  
  
!  
control-plane  
  
!  
line con 0  
    transport input none  
    stopbits 1  
line aux 0  
    stopbits 1  
line vty 0 4  
    login  
End
```

R5:

Building configuration...

Current configuration : 3907 bytes

```
!  
! Last configuration change at 23:36:51 UTC Tue Sep 17 2024  
!  
version 16.9  
service timestamps debug datetime msec  
service timestamps log datetime msec  
platform qfp utilization monitor load 80  
platform punt-keepalive disable-kernel-core  
!
```

hostname R5

!

boot-start-marker

boot-end-marker

!

vrf definition Mgmt-intf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

no aaa new-model

!

login on-success log

!

subscriber templating

vtp domain cisco

vtp mode transparent

ipv6 unicast-routing

multilink bundle-name authenticated

!

crypto pki trustpoint TP-self-signed-4153289585

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-4153289585

revocation-check none

rsakeypair TP-self-signed-4153289585

!

crypto pki certificate chain TP-self-signed-4153289585

certificate self-signed 01

!

license udi pid ISR4321/K9 sn FDO214417Q4

no license smart enable

diagnostic bootup level minimal

!

spanning-tree extend system-id

!

redundancy

mode none

!

interface GigabitEthernet0/0/0

ip address 192.168.5.2 255.255.255.0

negotiation auto

ipv6 address 2001:DB8:ACAD:6::2/64

ipv6 enable

ipv6 ospf 1 area 2

!

interface GigabitEthernet0/0/1

ip address 192.168.6.1 255.255.255.0

negotiation auto

ipv6 address 2001:DB8:ACAD:7::1/64

ipv6 enable

ipv6 ospf 1 area 2


```
!  
interface Serial0/1/0  
  no ip address  
  shutdown  
!  
interface Serial0/1/1  
  no ip address  
  shutdown  
!  
interface GigabitEthernet0  
  vrf forwarding Mgmt-intf  
  no ip address  
  shutdown  
  negotiation auto  
!  
router ospf 1  
  router-id 5.5.5.5  
  network 192.168.5.0 0.0.0.255 area 2  
  network 192.168.6.0 0.0.0.255 area 2  
!  
ip forward-protocol nd  
ip http server  
ip http authentication local  
ip http secure-server  
ip tftp source-interface GigabitEthernet0  
!  
ipv6 router ospf 1
```

router-id 5.5.5.

```
!  
control-plane  
!  
line con 0  
  transport input none  
  stopbits 1  
line aux 0  
  stopbits 1  
line vty 0 4  
  login  
!  
End
```

R6:

Building configuration...

Current configuration : 1538 bytes

```
!  
! Last configuration change at 18:58:39 UTC Tue Sep 17 2024  
!  
version 16.9  
service timestamps debug datetime msec  
service timestamps log datetime msec  
platform qfp utilization monitor load 80  
platform punt-keepalive disable-kernel-core  
!
```

hostname R6

```
!  
boot-start-marker  
boot-end-marker  
!  
vrf definition Mgmt-intf  
!  
address-family ipv4  
exit-address-family  
!  
address-family ipv6  
exit-address-family  
!  
no aaa new-model  
!  
login on-success log  
!  
subscriber templating  
ipv6 unicast-routing  
multilink bundle-name authenticated  
!  
license udi pid ISR4321/K9 sn FDO215009QY  
no license smart enable  
diagnostic bootup level minimal  
!  
spanning-tree extend system-id  
!  
redundancy
```

mode none

!

interface GigabitEthernet0/0/0

ip address 192.168.5.1 255.255.255.0

negotiation auto

ipv6 enable

ipv6 ospf 1 area 2

!

interface GigabitEthernet0/0/1

ip address 192.168.4.2 255.255.255.0

negotiation auto

ipv6 enable

ipv6 ospf 1 area 2

!

interface Serial0/1/0

no ip address

!

interface Serial0/1/1

no ip address

!

interface GigabitEthernet0

vrf forwarding Mgmt-intf

no ip address

negotiation auto

!

router ospf 1

router-id 6.6.6.6

network 192.168.4.0 0.0.0.255 area 2

network 192.168.5.0 0.0.0.255 area 2

!

ip forward-protocol nd

no ip http server

ip http secure-server

ip tftp source-interface GigabitEthernet0

!

ipv6 router ospf 1

router-id 4.4.4.4

!

control-plane

!

line con 0

transport input none

stopbits 1

line aux 0

stopbits 1

line vty 0 4

login

!

End

S1:

Building configuration...

Current configuration : 2560 bytes

!

```
version 12.2
no service pad
service timestamps debug uptime
service timestamps log uptime
no service password-encryption
!
hostname LAYER3SWITCH
!
boot-start-marker
boot-end-marker
!
no aaa new-model
system mtu routing 1500
ip routing
!
ipv6 unicast-routing
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
interface FastEthernet0/1
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/2
```

switchport access vlan 10

switchport mode access

!

interface FastEthernet0/3

!

interface FastEthernet0/4

!

interface FastEthernet0/5

!

interface FastEthernet0/6

!

interface FastEthernet0/7

switchport mode access

!

interface FastEthernet0/8

!

interface FastEthernet0/9

!

interface FastEthernet0/10

!

interface GigabitEthernet0/1

!

interface GigabitEthernet0/2

!

interface GigabitEthernet0/3

!

interface GigabitEthernet0/4

```
!  
interface Vlan1  
!  
interface Vlan10  
  ip address 192.168.0.1 255.255.255.0  
  ipv6 address 2001:DB8:ACAD:1::1/64  
  ipv6 ospf 1 area 1  
!  
router ospf 1  
  log-adjacency-changes  
  network 192.168.0.0 0.0.0.255 area 1  
!  
ip classless  
ip http server  
ip http secure-server  
!  
ipv6 router ospf 1  
  router-id 10.10.10.10  
  log-adjacency-changes  
!  
tftp-server flash:c3560-ipservicesk9-mz.SE7  
tftp-server flash:c3560-ipservicesk9-mz.SE7 alias ipservices  
!  
line con 0  
line vty 0 4  
  login  
line vty 5 15
```


login

!

End

Extra General Commands and/or Concepts:

Pings (ping) - creates a packet that is sent from a source to a destination; checks for connectivity

Trace route (tracert)- Creates a packet that is sent from a source to a destination and tracks its path towards the destination.

show ip ospf neighbors (v4 and v6) - Checks what adjacencies OSPF has managed to create with neighboring routers to see check whether it is setup correctly or not.

show ip ospf interface - shows the OSPF interface with its area, state, and configuration information. Can choose to enter an interface to see that specific process.

Topology - What the lab would look like, draw on paper or made in packet tracer.

ip route tables (v4 and v6) - A table of ip addresses created before the actual start of the lab to make setting ip addresses simpler and have less mistakes.

"enable ipv6" on Vlan (switch) - allows the Vlan to use process and use ipv6 addresses.

"ipv6 unicast-routing" in global config on switch - enables ipv6 traffic on a layer 3 switch

"switchport mode access" and "switchport access vlan x" on switch interfaces -

Assigns an interface to a Vlan

"ipv6 ospf x area y" on any interface that is connected through OSPFv3 - starts the IPv6 OSPF routing process on a VLAN.

set router-id for all ospf (v2 and v3) - gives each router an ID to identify each other.

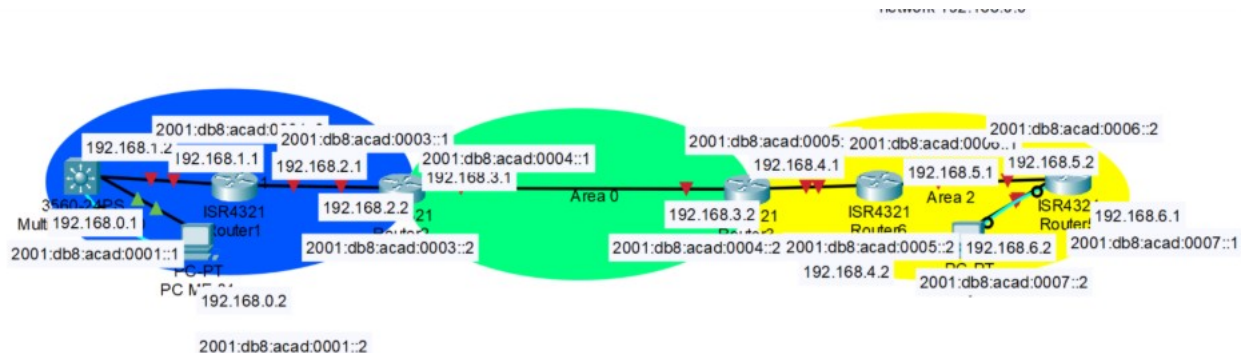
router ospf x - enable OSPF on routers/switch\

OSPF network statements (network xxx.xxx.xxx.xxx) – Advertises addresses all connected routers through OSPF

"ip add (xxx.xxx.xxx.xxx xxx.xxx.xxx.xxx)" or "ipv6 add (xxxx::x/xx)" – Sets an interface with an ipv4 or ipv6 address

"ipv6 router ospf" to enter/enable OSPFv3 (ipv6) and set ipv6 network statements for them as well.

Lab Diagram



Problems

The routers were not able to route to the switch or ping any interface on it due to a different method, which assigned two ports into a vlan and configuring the vlan, and that caused some confusion and issues, but was ultimately solved.

Conclusion

This lab is a great review on old topics that could have started fading in students' memory and gives them an idea about the difficulty and timeframe of future labs.